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CERN — EP-CMS-TDQ Group

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Application for Software Developer — CMS Trigger and Timing Control and Distribution System (EP-CMS-TDQ-2026-153-GRAP)

I first met physics in prépa. Two years of MPSI/MP at Lycée Saint-Louis taught me mechanics, electromagnetism, and the habit of working a problem until the mathematics agrees with the world. I then took an unusual path: a business degree across three continents, and a deliberate turn back to what I actually loved, teaching myself software engineering until Amazon hired me as a software development engineer. Working on the systems that keep the LHC's data flowing would close a loop that started in those classrooms, and it is the kind of engineering I want my career to serve.

My daily work maps closely to what the TCDS software effort needs. At Amazon I build and operate backend services in production: I design control and configuration paths, wire up monitoring and alerting, and carry the on-call pager for what I ship. This year I delivered a migration workstream that moved four pricing services off a legacy region, and re-platformed fee computation onto a new centralized engine. The part of that work I am proudest of is the validation method: we ran the new implementation in shadow against the legacy system at production scale and compared outputs before any cutover. That discipline of testing a replacement system at progressively larger scales before trusting it is, as I understand it, exactly the road the Phase-2 DAQ system will travel between now and 2028.

Collaborative development and CI/CD are second nature to me. I work in a large shared codebase with mandatory code review, automated test gates, and continuous deployment pipelines that I have built and maintained myself. I write design documents that compare options explicitly and present them to mixed audiences, and I have coordinated implementation standards across several engineering teams. On the interface side, I have shipped user-facing web components in TypeScript and Angular, and REST-based interfaces between services.

I want to be straightforward about where I would need to grow. My strongest languages today are Python, Java, and TypeScript; my C++ dates from coursework at UC Berkeley, and bringing it to modern, professional standard would be my first priority, starting before January. I have also not worked against custom electronics before. What I can offer against that is a track record of closing exactly this kind of gap: I learned software engineering itself this way, and every role I have held began with a domain I had to absorb quickly, from European pricing law to cross-region cloud networking.

Two practical notes. I am a French national, so beyond fluent English I bring native French to a Geneva-based team. And the timeline suits me: I am available for a January 2027 start, at the beginning of the commissioning years where sustained, careful engineering matters most.

I would be glad to discuss how I can contribute to the TCDS software and to the Phase-2 DAQ commissioning. Thank you for your consideration.

Sincerely,

Zoha Rakotomalala